

# Q4. Zepto Support Ticket Manager

DigiPlus IT Agentic AI Hackathon · 6-hour build · Thakur College of Engineering & Technology

## 1. Challenge

A 10-minute-delivery company gets thousands of support tickets a day — "items missing", "order late", "wrong item", "refund not received". Ninety percent are near-identical to tickets already resolved hundreds of times before, yet each one still waits for a human to apply the same fix as last time.

## 2. Problem Statement

Build a resolution system that learns from history: for each new ticket, find the most similar past resolved tickets, auto-resolve the same way when confidence is high (refund / redelivery / coupon), and queue for human review when it is not. The customer gets a drafted reply either way.

## 3. Dataset

| File                 | Contents                                                                       |
|----------------------|--------------------------------------------------------------------------------|
| resolved_tickets.csv | Historical resolved tickets: description, action taken, resolution note, CSAT. |
| new_tickets.csv      | Incoming tickets: description and linked order.                                |
| orders_context.csv   | Order facts: items, value, delivery time, status.                              |

## 4. Reference Flow (User Perspective)

Today — the pain:

1. "Milk packet missing" — the 4,000th ticket today — waits in a queue for hours.
2. An agent reads it, recognises it instantly, refunds Rs 40 — a decision history already knew.
3. Multiplied by thousands, daily; unusual tickets wait just as long behind routine ones.

With your system:

1. A new ticket is matched to its most similar past resolved tickets in seconds.
2. Routine tickets are auto-resolved the way history resolved them, with a drafted reply.
3. Unusual tickets land in a human lane with precedents and a suggested action attached.
4. The board shows what was auto-resolved, why, and with what confidence.

## 5. Expected Components in the Solution

| Component  | Expectation                                                                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Backend    | Similarity search over resolved tickets (TF-IDF is fine); confidence threshold decides auto vs human; simulated refund/redeliver/coupon actions; decisions logged. |
| Frontend   | Two-lane board (auto-resolved / needs-human); each ticket shows top-3 similar past cases, chosen action, confidence, and drafted reply.                            |
| AI layer   | Decide act-vs-escalate with an explicit confidence score; generate the reply; answer "why this action?" with the precedent tickets.                                |
| Deployment | Live public URL (free tier) + public GitHub repo.                                                                                                                  |
| Bonus      | Human approve/override controls with logging; live ticket-stream simulation; embeddings instead of TF-IDF.                                                         |

Build the core completely before touching anything in the Bonus row. A small finished system beats a large broken one.

## 6. Validation Scenarios — How an Ideal System Behaves

- A clear missing-item ticket with strong precedents is auto-resolved with the same action, a refund no larger than the order value, and a reply citing its top-3 precedents.
- A novel ticket with low similarity goes to the human lane — the system never acts on weak evidence.
- When top precedents disagree on the action, the ticket is queued, not guessed.

- A ticket on a cancelled order never triggers redelivery — order context constrains actions.